# Daniel Peelen

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## Education

**University of Pittsburgh** September 2023 - May 2027

*B.S. Computer Science, Hungarian Minor — 3.9 GPA, Dean's List*

- Coursework: Deep Learning, Data-Driven Systems, Data Structures & Algorithms, Systems Software

## Experience

**Microsoft** May 2026 – August 2026

*Software Engineering Intern* *Redmond, WA*

- Shipped an **org-wide plug-in for Microsoft's internal Copilot**, reducing engineers' triage times by **60%**.
- Analyzed **1,000,000+ Word crashes daily** with Kusto and OpenTelemetry by building a reliability AI agent.
- Fixed Word bugs surfaced by agent's output, **eliminated 50,000+ enterprise-facing crashes over 2 releases**.

**Synopsys** January 2026 – April 2026

*Software Engineer Intern* *Pittsburgh, PA*

- Designed and implemented Ansys Mechanical's RAG coding agent memory solution with semantic clustering.
- Built chatbot's full test suite and a multi-turn eval harness with bucket visualization on Ansys test infrastructure.
- **Raised multi-turn accuracy from 40% to 82%** by shipping Go code to production behind feature flags.

**Thursday (Carnegie Mellon startup)** June 2025 – December 2025

*Machine Learning Engineer* *Pittsburgh, PA*

- Shipped Monolith-based recommender to **10K monthly users** on Kafka, PyFlink, and Redis, serving under 200ms
- Built a thread-safe Cuckoo hash feature and embedding store from scratch, feeding a PyTorch DeepFM ranker.

**PNC Financial Services** May 2025 – August 2025

*Software Engineer Intern* *Pittsburgh, PA*

- Built a full-stack **housing analytics app** that turned **1M+ housing records** into homeowner-facing insights.
- Authored Angular unit tests, raising coverage from 45% to 85% and cutting production bugs by 35%.

## Projects

**DannyTorch** | *Python, NumPy*

- Built a deep learning framework in pure NumPy **without AI** with reverse-mode autograd over a dynamic graph
- Hand-wrote a decoder-only transformer and trained a 604K-param LM on it, **cutting loss from 4.9 to 1.2**

**GhostPIN** | *Node.js, TypeScript, Redis, PostgreSQL, WebAuthn* **2nd Place, HackHarvard 2025**

- Designed an anti-phishing SDK with RSA-signed protocol and origin-bound nonces that block fake checkout pages.
- Integrated biometric MFA into checkout flow via WebAuthn, with approvals verified end-to-end in under 200ms.

## Leadership

**University of Pittsburgh** August 2024 – April 2025

*Teaching Assistant* *Pittsburgh, PA*

- Led recitations and practices for 100+ students in Discrete Math, **raised average final scores from 78 to 84.**
- Held 8+ hours of weekly office hours, coaching 50+ students on **proof techniques and algorithmic thinking**.

**Director** January 2025 – September 2025

*SteelHacks* *Pittsburgh, PA*

- Grew the event **21%** year over year to 600+ attendees and raised $50K+ from Google, Amazon, and PNC.

**Technical Team Leader** September 2026 – Present

*Google Developer Group, Budapest University of Technology and Economics* *Budapest, Hungary*

- Led team of engineers to build myBME for students and co-organize guest tech lectures **while studying abroad.**

## Technical Skills

**Languages**: Python, C/C++, Java, Go, TypeScript, JavaScript, SQL, HTML/CSS, OCaml
**Machine Learning & AI**: PyTorch, NumPy, HuggingFace, transformers, model evals, benchmarking
**Cloud & DevOps**: AWS, Google Cloud Platform, Azure DevOps, Docker, Linux, Git, CI/CD, Agile, Firebase
**Backend & Frontend**: Node.js, Flask, Spring Boot, Kafka, Redis, React, Angular, JUnit, Maven, Gradle